

24/08/2026

Monday

[Vrinda]

The selected datasets provide complementary coverage for the RoadSentinel pothole detection and road-health monitoring pipeline. Rather than relying on a single dataset, the collection combines pixel-level segmentation data, drone imagery, Indian road imagery, water-filled pothole examples, and depth/disparity information. This makes the overall dataset collection suitable for developing different components of the proposed system.

Pothole Mix is the strongest primary dataset for pothole segmentation because it contains 4,340 image/mask pairs with pixel-level binary segmentation annotations and predefined training, validation, and test splits. It is therefore well suited for developing and evaluating the pothole segmentation component. However, its images are not drone-based, so it does not directly represent the intended aerial viewpoint.

RDD2022 provides important domain diversity and is particularly valuable because its China_Drone subset contains genuine UAV imagery. The dataset includes bounding-box annotations for four road-damage categories, including potholes. The China_Drone subset contains 482 training images, while the India subset provides a larger collection of Indian road imagery. Since the annotations are bounding boxes rather than segmentation masks, SAM2 can be used to generate initial masks from the annotated pothole regions, followed by manual correction where required.

The **Annotated Water-Filled and Dry Potholes Dataset** is an important targeted supplement because it specifically contains water-filled, muddy-water, and visually difficult potholes. With 713 images and 1,152 manually annotated potholes, it can be used to improve robustness to water-filled potholes and support development of a dedicated water-pothole classification component.

Pothole-600 is particularly useful for the depth and severity components because it provides both pixel-level pothole masks and disparity/depth maps. Although it is smaller than the primary datasets and is vehicle-mounted rather than drone-based, it provides a valuable basis for investigating RGB-based depth estimation and evaluating depth-related methods.

The **Chitholian Pothole Detection Dataset** is comparatively small, containing 665 images and 1,740 labeled pothole instances. It is therefore more suitable for rapid prototyping, annotation-format testing, and smoke testing of the pipeline than for serving as the primary training dataset.

The **Multi-Weather Pothole Detection (MWPD)** dataset provides additional variation in weather and illumination conditions, including rainy, daytime, twilight, and nighttime scenarios. It can be used as supplementary data for improving robustness under varying environmental conditions, although several dataset statistics require verification before being

used as formal training or evaluation data.

The **QR4Change Urban Civic Issues Dataset** contains 1,004 pothole images and 1,962 plain-road negative images from Pune, India. Since it provides image-level classification rather than localization annotations, it is better suited for negative-sample enrichment and domain-specific classification rather than pothole segmentation or precise localization.

Overall, the dataset collection is strong because each dataset can serve a distinct role in the RoadSentinel pipeline. Pothole Mix can provide the primary segmentation supervision, RDD2022 can provide drone and Indian-road domain coverage, the Water-Filled/Dry dataset can address water-pothole robustness, and Pothole-600 can support depth-related experimentation.

The main limitation is the lack of a large public dataset containing **drone-native pothole images with pixel-level segmentation masks**. The China_Drone subset of RDD2022 provides genuine UAV imagery, but it contains only bounding-box annotations. Consequently, an important part of the proposed work will be adapting segmentation knowledge from ground-level datasets to the drone domain and, where necessary, creating corrected drone-specific masks using SAM2-assisted annotation.

Therefore, the datasets should not simply be merged into one undifferentiated training set. Instead, they should be assigned different roles within the RoadSentinel system, with dedicated training, validation, domain-adaptation, depth-evaluation, and robustness experiments performed according to each dataset's annotation type and imaging characteristics.

[Nitin]

The initial RoadSentinel pipeline was received from **Marion**, containing the core implementation for road-health analysis, memory-bank construction, DINOv2 feature extraction, pothole localization, depth/area estimation, and supporting modules.

The received implementation was reviewed and adapted to fit the current RoadSentinel project structure and deployment requirements. The code was organized into dedicated modules for **CARLA simulation, common utilities, inference, memory-bank generation, evaluation, and Raspberry Pi edge processing**. The pipeline was also connected to the project's existing dataset structure and the initial China_Drone development subset.

The modified pipeline now contains separate components for:

- **DINOv2 embedding extraction** for road-feature representation.
- **Healthy-road memory-bank construction and coreset selection**.

- **SAM2-based pothole mask generation.**
- **Pothole localization and area estimation.**
- **RGB-based depth estimation** and depth evaluation utilities.
- **GPS-based pothole localization.**
- **CARLA RGB/depth image capture and drone-control components.**
- **Raspberry Pi edge-processing and telemetry modules.**
- **Inference server and end-to-end inference execution.**
- Separate dependency files for **GPU development** and **Raspberry Pi deployment**.

The existing implementation was also placed under version control and uploaded to the RoadSentinel GitHub repository so that the project code can be maintained collaboratively and further developed.

The current 35-image China_Drone subset (25 healthy-road images and 10 test-road images) is being retained for development and smoke testing. The larger public datasets are currently being acquired and validated before being used for proper training and evaluation.

Improvements

[Vrinda]

24-8([DownloadDatasets.md](#)) , work to be done on the datasets includes instructions on how to push work to github as well.

[Nitin]

Need to work on the CARLA environment including UAV speed and height adjustments.
Verify Dataset Vrinda will give and do format conversion

[Marion]

Need to fine tune model, improve memory Bank
24-8(ImprovementToPipeline)